Quality Assurance 344

Welcome to the presentations of Quality Assurance 344.

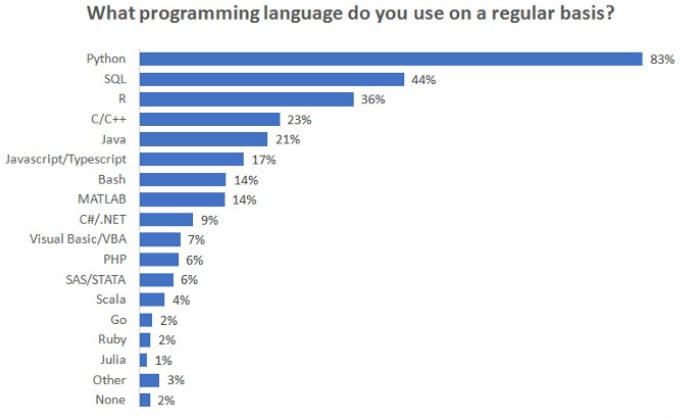
Your lecturer is: Theuns Dirkse-van Schalkwyk.

The course includes ECSA GA4 (Graduate Attribute). Please read through the module document for more information, since you must pass both the subject matter and the GA4 outcomes.

Make sure you have a working R and Rstudio setup on your computer.

With that, let’s get started...

Quality Assurance 344



Cannondale
Giant
Merida
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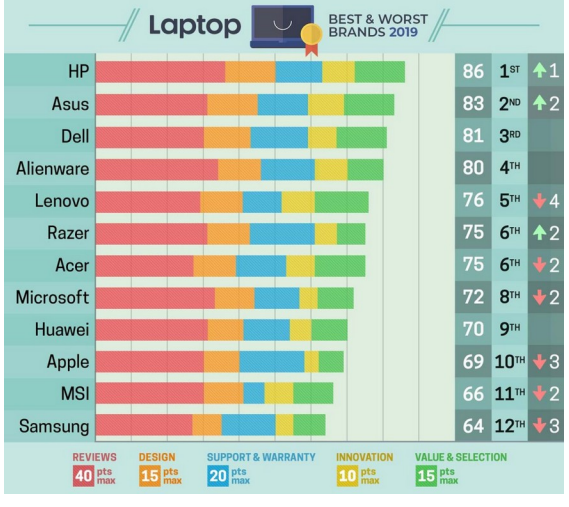
Scott
Santa Cruz
Specialized



Apple iPhone 11 PRO 256GB Gold
R 25 399,00 powerbite
And then there was Pro. A transformative triple-cam
iOS · 12 MP Rear Camera · 256 GB · Smartphone

Samsung Galaxy Note 8 64 GB Sm
R 49 950,00 Manicaa 69% positive (173)
Product Details Colour Name:Maple Gold †|† Style
Android · 8 MP Rear Camera · 4G LTE · 64 GB · S

Bartec PIXAVI Impact X ATEX, IEC
R 65 417,04 RS Components South Africa
Bartec PIXAVI Impact X ATEX, IECEx Black Smart
Android · 4.3 inches screen · 8 MP Rear Camera ·

Huawei Mate Xs 5G (512GB, Dua
R 67 995,00 Better Buys SA
Buy the Huawei Mate Xs 5G (Dual Sim) in Inters
Huawei · Mate · Android · 6.6 inches screen · 40

Q and q (read: big Q and little q)

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To assure that product quality (q) is good, companies have systems to research, plan, measure, analyse the data and make improvements. The systems to manage and improve product and service quality is called “big Q” (Q).

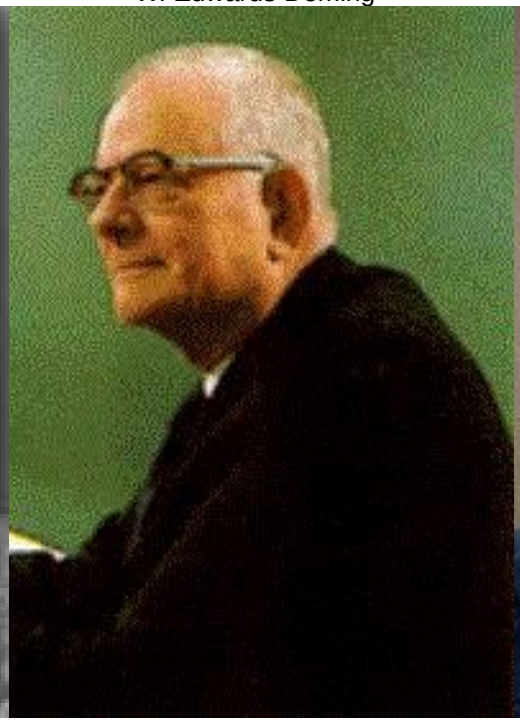
It is clear that companies improve q by ensuring their Q is well managed. Of course, to ensure clarity in the workplace we should rather say simply: ”Companies improve products and services by ensuring their quality systems are well managed.”

q

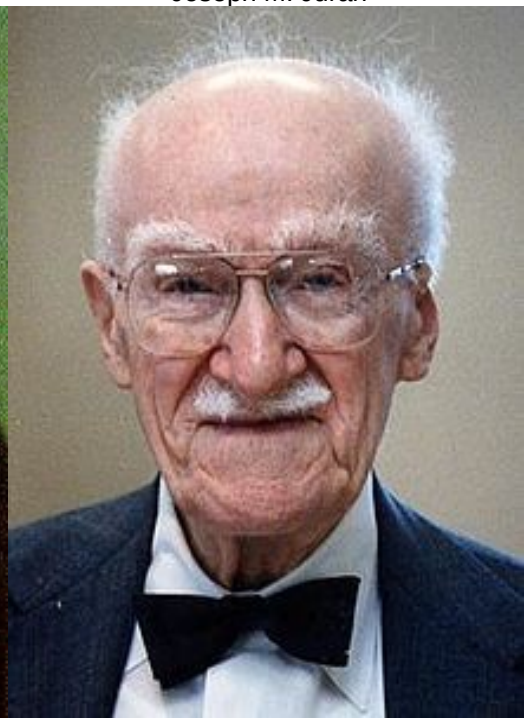
Q

Walter Shewhardt

W. Edwards Deming



Joseph M. Juran

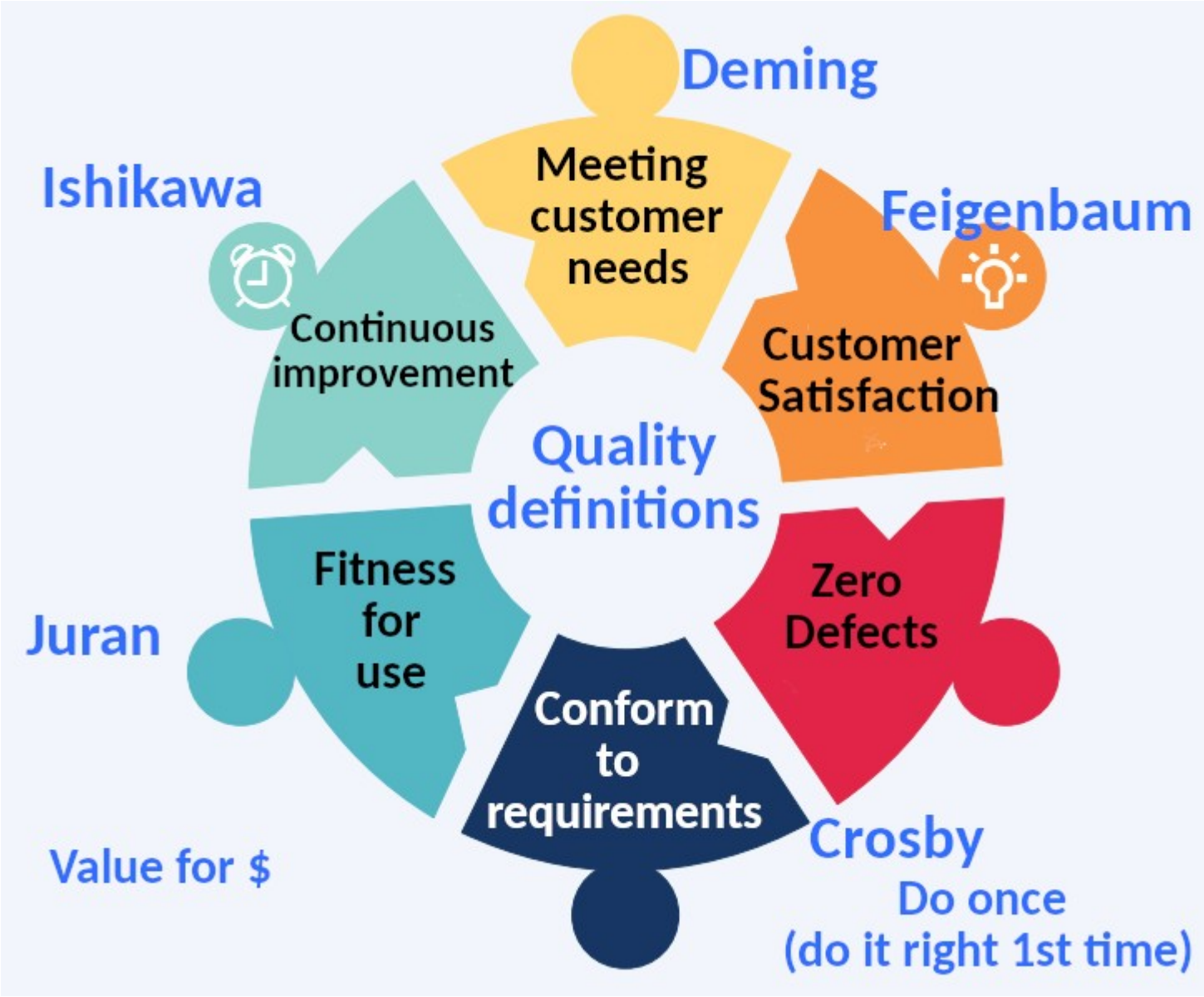


Armand Feigenbaum

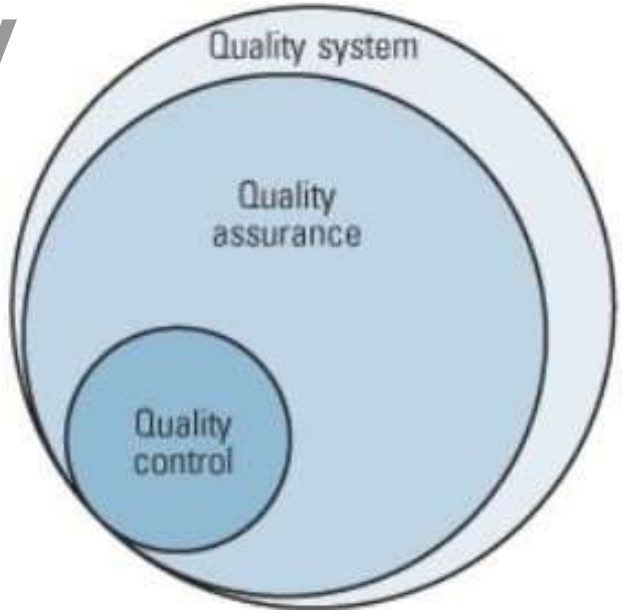
Philip Crosby

Kaoru Ishikawa

Quality defined



Assurance of Quality



- Assurance is a positive declaration intended to give confidence; a promise. To be able to give this assurance of quality, companies use Quality Management Systems.
- In these systems Quality Assurance looks at providing proof to managers that their system is performing as it was designed and products and services are meeting the requirements of customers.
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Organizations must adopt an interdisciplinary series of quality controls to achieve these principles. The nine core elements of a QMS should include a people focus, quality objectives, a quality manual, organizational responsibilities, data management, and other practices.

Who developed the quality systems and tools we use today?

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When people ponder “Quality” they have many different perspectives. PRODUCT Quality may be seen as top brands: NIKE, PORSCHE, APPLE, GUCCI or value for money: TOYOTA, SAMSUNG, WOOLWORTH'S.

In fact each USER of products have their own idea of what quality means for them.

For workers who MANUFACTURE products, quality means that their output must conform to certain standards.

Similarly, in the service industry, quality is seen as services that meet the requirements of the users.

From the previous discussion it seems that “Quality” is generally perceived as the degree of excellence of products or services, relative to what users require from them.

But for “Quality Assurance”, product quality is only one half of the coin.

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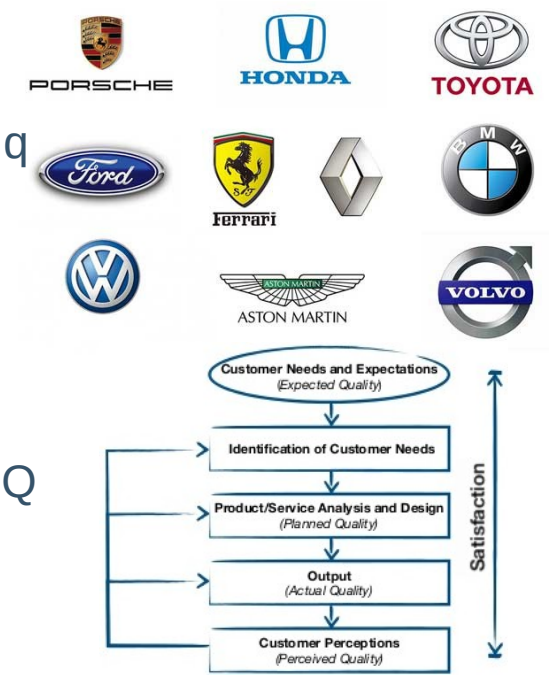

In every field of study, experts use specific words that have a unique meaning or application to their work. For instance in the sport Rugby, the word "try" has a different meaning than when I say: "I'll try harder!"

When you enter into a new field of study, learning these words and their specific meanings are just as important as getting the "big picture".

In Quality management big Q and little q are two such words.

As we've seen previously, most people and customers only care about the quality they experience in their products or services they buy. In quality management the quality of products or services is called "little q" (q). q is the quality embedded or encapsulated in the product or service. It will become clear through this course that q depends on understanding the needs and the requirements of the customer.

Companies also care about q, since customers vote with their dollars. If q is bad, they leave and take their dollars with them.

To assure that product quality (q) is good, companies have systems to research, plan, measure, analyse the data and make improvements. The systems to manage and improve product and service quality is called "big Q" (Q).

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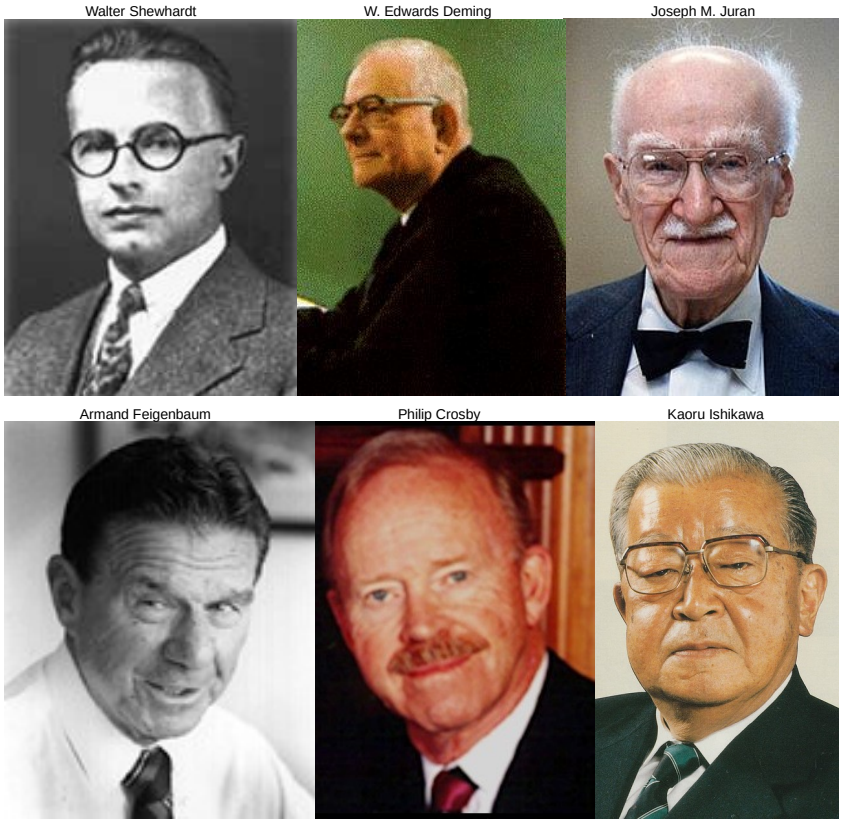


Many of our quality definitions come from one of the quality gurus, such as Deming, Juran, Crosby, Ishikawa and Feigenbaum. However, they were all standing on the shoulders of another giant, Walter Shewhardt. Both W. Edwards Deming and Joseph M. Juran worked with Shewhardt and popularised and built on top of his quality foundations.

Shewhardt contributed two huge concepts that were not improved on for about 40 years and is even still in use today.

Regarding product quality (q) Shewhardt developed the idea of using statistical control charts to monitor processes and empower workers to make informed decisions about control of their processes. We'll delve deeper into this in the latter part of this course.

Regarding the Systems to manage quality (Q) Shewhardt proposed the Plan, do, check and act cycle, that is at the heart of any quality system and drives continuous improvement.

After the second world war, manufacturing in the United States was mostly happy to continue with their "status quo" as the world leader, while Japan was hungry to repair their economy and growth. As one of the early leaders in the quality movement, Deming was invited in 1951 to assist Japan in this mission, training the leaders of Japanese industry. Many in Japan credit Deming as one of the inspirations for what has become known as the Japanese post-war economic miracle of 1950 to 1960, when Japan became the second-largest economy in the world through processes influenced by the ideas of Deming. Deming made a significant contribution to Japan's reputation for innovative, high-quality products, and for its economic power. He is regarded as having had more impact on Japanese manufacturing and business than any other individual not of Japanese heritage.

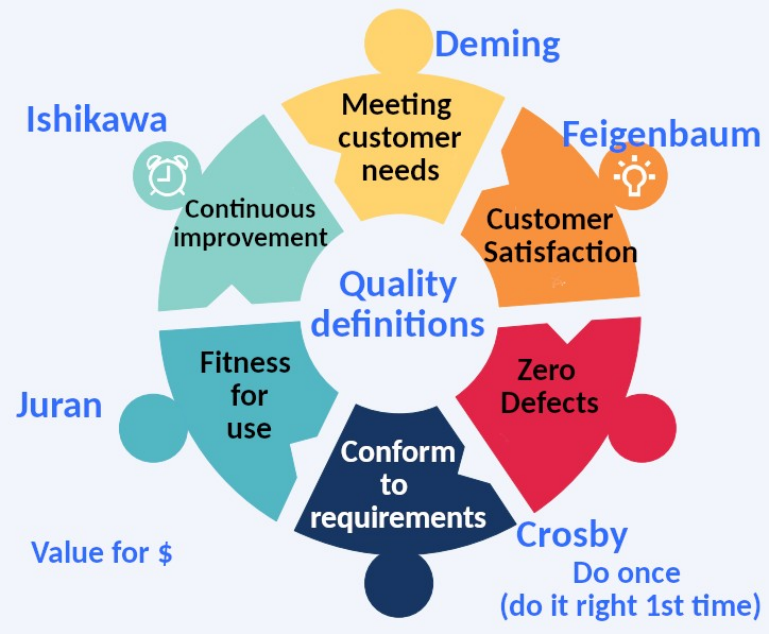
Joseph M. Juran, while working at Hawthrone Works in 1925 attended training in the methods of statistical sampling and control chart techniques developed by Shewhardt. This basically launched his career in quality management. He was also contracted by Japan to help with the development of their managers during 1954-1960.

Armand V. Feigenbaum developed Total Quality Management (TQM) during the 1940's.

Philip B. Crosby developed management theory and quality management practices such as Zero Defects in the 1970's.

Kaoru Ishikawa was instrumental in convincing Japanese managers to listen and apply the concepts of Deming that was largely responsible for Japan's quality-improvement initiatives. He translated, integrated and expanded the management concepts of Deming and Juran into the Japanese system. In 1962 he developed the concept of quality circles and in 1982 the Ishikawa diagram.

Quality defined



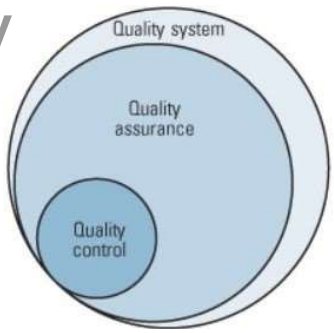
In general, we may say quality is the degree of excellence in products or services. But as we've seen q and Q makes defining quality in companies more complex. Thus, there are many different concepts and ideas to develop further in the quest for quality.

Some common themes are visible in the definitions we see.

1) Quality is based on user requirements or needs being satisfied.
2) Quality improves through repeated effort and is not static. The system should force us to do continuous improvement.
3) Through repeated efforts we should be able to get to a point where we do it right the first time; ie zero defects.

It is important to realise that there is no one perfect definition. Quality is context aware.

Assurance of Quality



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Organizations must adopt an interdisciplinary series of quality controls to achieve these principles. The nine core elements of a QMS should include a people focus, quality objectives, a quality manual, organizational responsibilities, data management, and other practices.

Now that we have some understanding of both product quality and the management of quality, what is assurance? Assurance is defined as: a positive declaration intended to give confidence; a promise.

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For week 1, choose 5 quotes by at least 3 different authors from the list.

Write 150-400 words on how these quotes fit into the concepts of Q, q and process control.